American Computer Science League

2020-2021 ● Contest 2: Solutions ● Elementary Division

1. Prefix / Infix / Postfix

5 3 + 4 2 − * 7 1 / + 9 + 8 /
= ((5 + 3) * (4 − 2) + (7 / 1) + 9) / 8 = (8 * 2 + 7 + 9) / 8
= (16 + 16) / 8 = 32 / 8 = **4**

1. 4 (A)

2. Prefix / Infix / Postfix

− + / 6 3 ^ * 7 1 2 / + + 8 9 4 7
= − + (6 / 3) ^ (7 * 1) 2 / + (8 + 9) 4 7
= − + (6 / 3) ((7 * 1) ^ 2) / ((8 + 9) + 4) 7
= − ((6 / 3) + ((7 * 1) ^ 2)) ((8 + 9) + 4) / 7)
= (2 + 7 ^ 2) − (21 / 7) = 51 − 3 = **48**

2. 48 (C)

3. Prefix / Infix / Postfix

((− 5 1) * (+ 4 8)) / ((^ 7 2) − (^ 5 2))
= (* − 5 1 + 4 8) / (− ^ 7 2 ^ 5 2)
= **/ * − 5 1 + 4 8 − ^ 7 2 ^ 5 2**

3. / * − 5 1 + 4 8 − ^ 7 2 ^ 5 2 (D)

4. Prefix / Infix / Postfix

((7 − 1 * 5 + 9 / 3) ^ 2 − 7) / 3
= ((7 − (1 5 *) + (9 3 /)) 2 ^ 7 −) / 3
= (7 1 5 * − 9 3 / +) 2 ^ 7 − 3 /
= **7 1 5 * − 9 3 / + 2 ^ 7 − 3 /**

4. 7 1 5 * − 9 3 / + 2 ^ 7 − 3 / (C)

5. Prefix / Infix / Postfix

8 4 * 5 1 / + 5 2 ^ − 2 3 * /
= (* 8 4) (/ 5 1) + (^ 5 2) − (* 2 3) /
= (− (+ * 8 4 / 5 1) ^ 5 2) (* 2 3) /
= **/ − + * 8 4 / 5 1 ^ 5 2 * 2 3**

5. / − + * 8 4 / 5 1 ^ 5 2 * 2 3 (E)